JEAN-ERIC TSANGA

ELECTRICAL & COMPUTER ENGINEER

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PROFILE

Highly motivated and detail-oriented Electrical and Computer Engineer with a strong foundation in designing, analyzing, and optimizing complex electrical systems and embedded technologies. Skilled in integrating hardware and software to create innovative and efficient solutions for modern engineering challenges.

EXPERIENCE

Web Development Trainee — **Jan 2024 – Apr 2024**
The Global Tech Experience · Remote

Amazon — **Apr 2022 – Present**

Electrical Engineer Teacher — **Sep 2014 – Sep 2021**
Ministry of Secondary Education - Cameroon

EDUCATION

Bachelor of Electrical and Computer Engineering — **Expected May 2027**
University of Arizona - Senior student

Bachelor's in Electrical Engineering — **Sep 2009 – Jul 2014**
University of Douala - Cameroon

PROJECTS

Amazon-Style E-Commerce Application — View live site

- Built a full-stack, Amazon-inspired shopping site with product browsing and details, search and filtering, a persistent (local-storage) cart, checkout, and responsive desktop and mobile layouts.
- JavaScript frontend integrated with a Node.js/Express backend and MongoDB, with user authentication and RESTful API integration.

Technologies: HTML, CSS, JavaScript, Bootstrap, Node.js, Express.js, MongoDB, REST APIs, Git/GitHub

Blood Loss Estimation Device

- Real-time blood-loss quantification device for surgical and obstetric settings: hemoglobin is extracted from used gauze into a fixed-volume reagent (Drabkin's) and its absorbance measured at 540 nm using the Beer–Lambert law.
- Disposable cartridge (sieve, reagent chamber, optical cuvette) with dual-wavelength LEDs (540 nm and 850–880 nm) and an OPT101 photodetector; a microcontroller corrects for turbidity using the IR channel and estimates blood loss in mL from the patient's baseline hemoglobin.

Technologies: microcontroller, Optical absorbance sensing, OPT101 photodetector, dual-wavelength LEDs, Beer–Lambert law

Audio Power Amplifier

- Audio power amplifier circuit project: Design a PCB using EAGLE Software

Research: How Much Branch Prediction Does an In-Order RISC-V Core Need?

Technologies: .RISC-V (RV64IMAFD) .Rocket Chip core generator · Chipyard SoC framework · Scala · Docker / VS Code Dev Containers

TECHNICAL SKILLS

Programming: . C · C++ · Python · Java · JavaScript · HTML · CSS · MIPS assembly
Hardware & HDL: . Verilog · VHDL · FPGA programming · Circuit design · Embedded Systems . PCB Design

LANGUAGES

.English · French

HOBBIES

Traveling · Music · Sport

CERTIFICATION

Web design and web development